

RxAuth AI

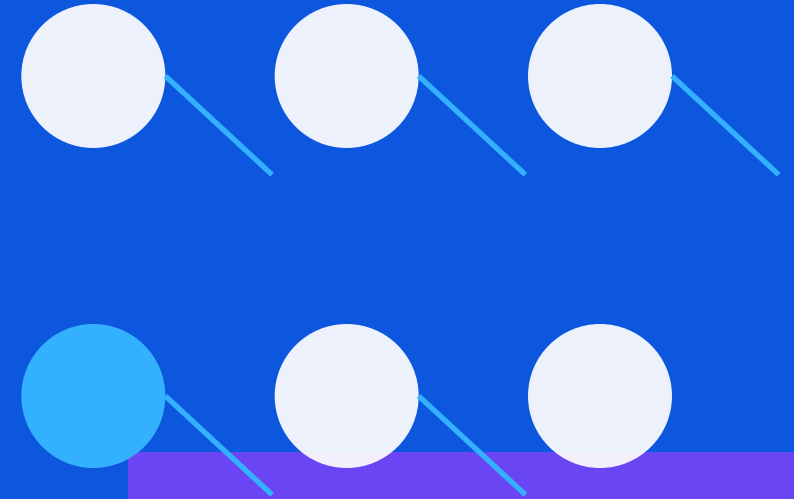
Evidence-grounded prior authorization intelligence

A human-in-the-loop AI engineering prototype for specialty pharmacy workflows

Bavely S. Tawfik

Python • scikit-learn • OpenCV • Pydantic • reproducible evaluation

August 2026



Executive summary

One project demonstrating the complete AI engineering lifecycle

01

The problem

Prior-authorization packets are document-heavy, policy-dependent, and vulnerable to missing or ambiguous evidence.

02

The solution

Ingest and classify documents, match patient evidence to policy criteria, expose gaps, and enforce provenance before review.

03

Current proof

Phase 1.5 includes typed ingestion, a leakage-resistant ML baseline, an end-to-end case spine, tests, and reports.

PROJECT THESIS

Use probabilistic AI for interpretation; use deterministic code for explicit rules; route uncertainty to a human.

Why prior authorization needs intelligence, not just automation

The bottleneck is evidence completeness and traceability before submission



Fragmented inputs

Clinical notes, labs, prescriptions, referrals, insurance cards, histories, forms, and miscellaneous correspondence.

Policy translation

Natural-language requirements must become explicit checks while preserving payer, version, page, and source text.

Trust gap

A confident-sounding answer is not enough; reviewers need evidence, citations, missing-data flags, and calibrated uncertainty.

Scope and safety guardrails

The portfolio version is administrative decision-support using synthetic data only

IN SCOPE

- Prepare a case for human review
- Classify and organize synthetic documents
- Match evidence to explicit policy criteria
- Flag missing, ambiguous, or low-confidence evidence
- Preserve source provenance end to end

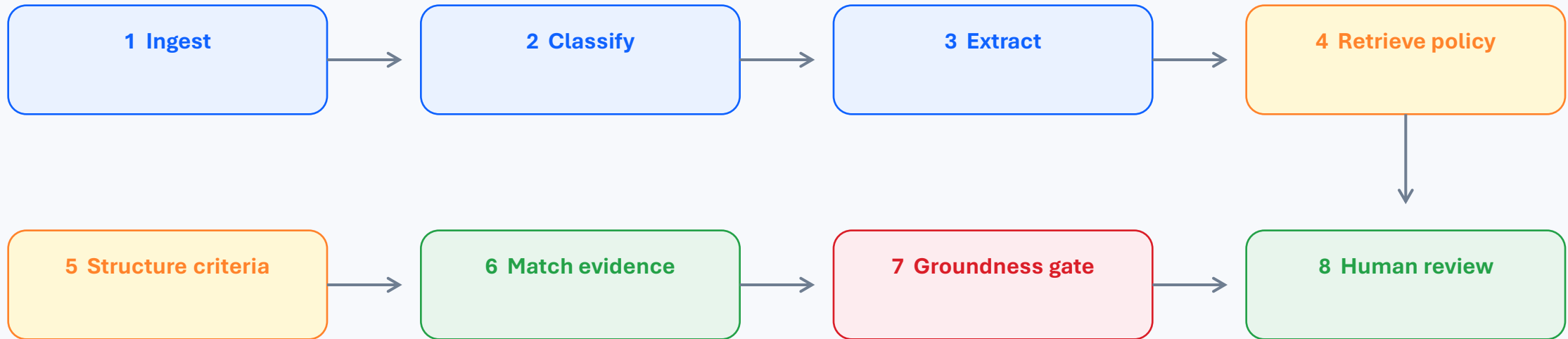
OUT OF SCOPE

- Diagnose, prescribe, or recommend treatment
- Approve or deny medical necessity
- Use real PHI in the public portfolio
- Infer live member benefit status from policy text
- Submit autonomously to a payer

Human review is mandatory: model output never becomes submission-ready by itself.

End-to-end AI workflow

A linear reviewer-facing flow with controlled intelligence inside each stage



IMPLEMENTED NOW

Typed ingestion + classification + deterministic matching + provenance gate

Typed entities make provenance enforceable

Important state lives in validated structures, not only inside prompts

Case

payer • medication • indication • documents

Document

type • confidence • page count

Evidence

normalized value • confidence • source span

Policy

version • effective date • criteria

Criterion

operator • expected value • policy source

Evaluation

five-state result • evidence IDs • explanation

Provenance contract: document + page + source text + confidence + method

From files to page-level text

One typed contract for text, text-bearing PDF, and scanned-image paths



DOCUMENT INTAKE
Patient ID: SYNTH-0001. Requested medication: Drug C. | Document received and processed on file. | Progress note for patient SYNTH-0001 seen by Dr. A. Rivera. | On file with Placeholder Insurance Co. | No acute distress noted on exam today.
End of intake record

SPECIALTY WORKFLOW RECORD
Regarding patient SYNTH-0002
ALT: 7.5/8 collected 2025-08-22.
Results reviewed and released to ordering provider.
Document received and processed on file.
Confidential synthetic example

32

Rendered benchmark

16 text-bearing PDFs + 16 scan-like images. Degradations include clean, rotated, blurred, low-contrast, and noisy variants.

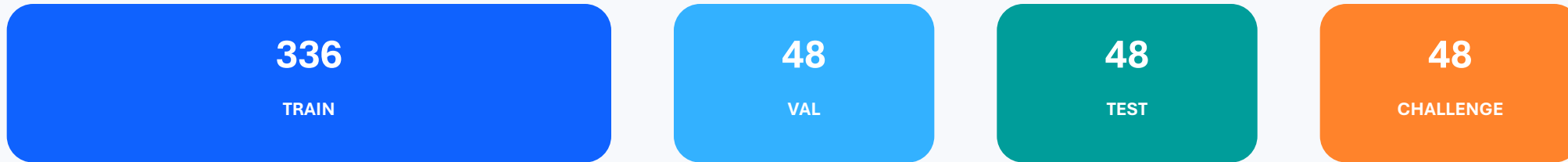
0.000

Measured honestly

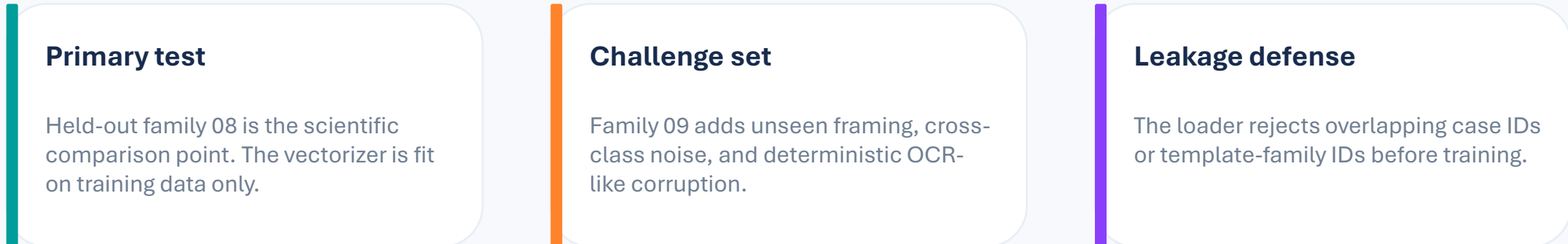
PDF mean character error rate: 0.000. Image preprocessing: 100%. OCR character accuracy: not reported without a configured runtime.

Synthetic dataset with leakage-resistant splits

Cases and template families are mutually exclusive across partitions

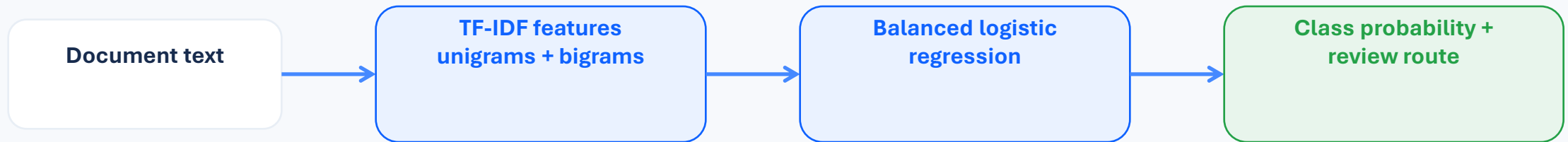


480 documents • 8 balanced classes • 10 template families • deterministic seed 42



Classical baseline: TF-IDF + logistic regression

A fast, interpretable benchmark before introducing a transformer



Feature design

Lowercase English text, stop-word removal, 1–2 grams, minimum document frequency 2, sublinear term frequency.

Training design

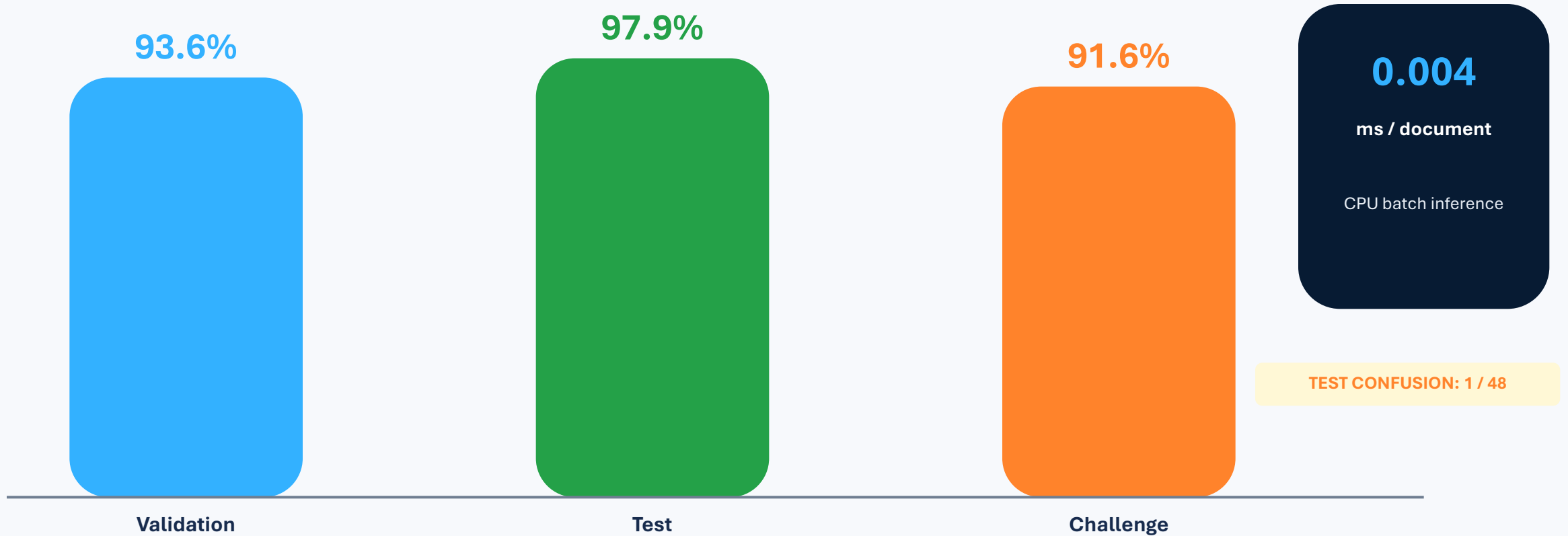
Balanced class weights, maximum 1,000 iterations, random state 42, training split only.

Decision policy

A 0.65 confidence threshold routes uncertain predictions to human review; it is not clinically validated.

Classifier performance: strong test, harder challenge

Macro F1 is the primary balanced metric; all values come from reproducible reports



Accuracy is not the same as confidence quality

Calibration and review-routing reveal the next engineering priority

97.9%

Test accuracy

Correct class prediction on the grouped held-out test set.

56.1%

Mean confidence

Average maximum probability despite strong accuracy.

0.418

Calibration error

10-bin expected calibration error; lower is better.

68.8%

Review routing

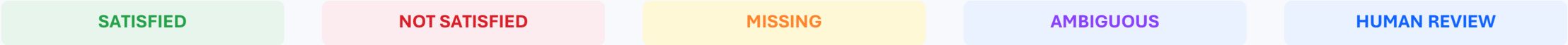
Predictions below the 0.65 threshold.

Engineering interpretation

The classifier is often correct but under-confident. Probability calibration and threshold selection matter more next than chasing a higher headline accuracy.

Criteria-to-evidence matching uses a five-state contract

Deterministic rules first; ambiguity is represented instead of guessed



One synthetic case, six policy criteria

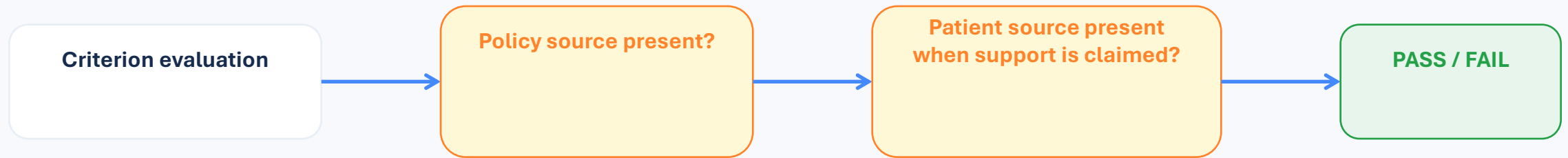
PA-DEMO-001 proves the workflow spine and reviewer-ready traceability



ID	Policy requirement	Result	Patient evidence
C1	Diagnosis documented	SATISFIED	clinical_note.pdf p.1
C2	≥ 12 weeks prior therapy	SATISFIED	medication_history.pdf p.2
C3	Inadequate response	SATISFIED	medication_history.pdf p.2
C4	A1c below 8.0	SATISFIED	lab_report.pdf p.1
C5	Screening document	MISSING	No evidence found
C6	Recent course du	GROUNDEDNESS GATE: PASS	
			“several months”

Groundedness gate before reviewer display

No supported claim may exist without patient evidence and a policy source



Current gate

Structural groundedness: validates evidence IDs, patient provenance, and policy provenance for each result.

Future gate

Semantic claim verification for generated text, citation correctness, conflicting evidence, and unsupported-claim rate.

Human control

Missing or ambiguous data becomes a visible review task. No autonomous submission is allowed.

Reproducible, testable, and honest by design

The repository is organized as an installable Python 3.12 package

21 PASS

Automated tests

Pipeline states, policy mismatches, unit conflicts, classifier persistence, split isolation, rendering, and ingestion.

seed 42

Reproducibility

Locked environment, deterministic seed, generated manifests, CLI entry points, and versionable report artifacts.

3 reports

Evaluation artifacts

Classifier report, ingestion report, structured end-to-end case JSON, and persisted trusted model bundle.

- Vectorizer fit on training split only; challenge set excluded from model selection
- Errors and calibration are reported alongside headline performance
- Synthetic-only scope and unmeasured OCR accuracy are stated explicitly

How RxAuth AI maps to IBM AI Engineering learning goals

Each course block unlocks a concrete project capability and measurable deliverable

IBM learning block	RxAuth AI deliverable	Status
Python / data	Typed ingestion, preprocessing, manifests	Implemented
Machine Learning	TF-IDF + logistic regression benchmark	Implemented
Deep Learning	Transformer classifier vs. same benchmark	Next
Computer Vision / NLP	OCR and evidence extraction with confidence	Partial / next
LLM applications + RAG	Public payer-policy retrieval and criteria extraction	Roadmap
AI agents	Controlled state graph and review checkpoints	Roadmap
Model evaluation	F1, calibration, routing, latency, groundedness	Implemented / expands

From Phase 1.5 to a portfolio-ready V1

Advance only when each layer has a defensible evaluation boundary



Model selection rule: same splits, same metrics, same reporting contract—then compare quality, calibration, latency, size, and deployment cost.

What this project demonstrates

Questions?

A course project can be technically rigorous, domain-aware, and honest about its limits

1 End-to-end thinking

Data contracts, preprocessing, training, evaluation, inference, decision logic, trust gates, and human review form one coherent system.

2 Evaluation maturity

Grouped splits, challenge data, calibration, review routing, latency, failure cases, and explicit non-claims show scientific discipline.

3 Responsible design

Synthetic data, administrative scope, source provenance, deterministic rules, and no autonomous submission reduce avoidable risk.

Build an AI system that can prove where every important answer came from.

Reproduce the demo and benchmarks

All commands run from the repository root

```
uv sync --group dev
uv run rxauth-build-dataset
uv run rxauth-benchmark-ingestion
uv run rxauth-train-classifier
uv run rxauth-milestone0
uv run pytest
```

Expected: 21 tests pass

Outputs: reports/*.md + case JSON

No real PHI

Evidence behind this presentation

Every metric and design claim maps to a repository artifact

README.md

Project thesis, scope, architecture, IBM course mapping, roadmap, and security posture.

docs/phase-1.5.md

Typed ingestion contract, grouped split design, current outcomes, and next step.

reports/classifier_baseline.md

Accuracy, macro F1, confidence, calibration, routing, latency, confusion matrix, failures.

reports/ingestion_benchmark.md

Rendered corpus size, PDF character error rate, preprocessing success, OCR non-claim.

reports/case_PA-DEMO-001.json

Six criteria, statuses, explanations, confidence, policy source, and patient evidence.

tests/

Executable checks for matching behavior, ingestion, dataset contracts, and classifier persistence.